

Tianchu Zeng

PhD Candidate, Electrical & Computer Engineering · National University of Singapore
Singapore · +65 8790 5814 · tianchuzeng@gmail.com

ORCID: 0009-0009-0773-4502 · [Google Scholar](#) · [GitHub](#) · [LinkedIn](#) · [Website](#)

PROFESSIONAL SUMMARY

PhD candidate in Electrical & Computer Engineering at the National University of Singapore researching reliable machine learning for complex scientific systems. Develops deep-learning methods for expensive biophysical brain models and rigorous statistical methodology for comparing biomedical ML algorithms under cross-validation. Combines research ownership with Python, PyTorch, JAX, HPC, Docker, and backend engineering experience.

EDUCATION

National University of Singapore — Singapore

Aug 2022 - Present (expected 2027)

Ph.D. in Electrical & Computer Engineering

- Advisor: Assoc Prof. BT Thomas Yeo
- Co-advisor: Prof. Thomas E. Nichols, University of Oxford
- Thesis: Machine Learning in Brain Imaging
- GPA: 4.88 / 5.0

Tsinghua University — Beijing, China

Aug 2018 - Jun 2022

B.Eng. in Department of Automation

- GPA: 3.74 / 4.00
- Thesis: Research on Trajectory Prediction Methods for Traffic Agents in Mixed Traffic Environments
- Advisor: Prof. Jianming Hu & Prof. Yi Zhang

RESEARCH EXPERIENCE

Machine Learning in Brain Imaging — National University of Singapore

Aug 2022 - Present

- Develop deep-learning methods for optimizing expensive, large-scale biophysical brain circuit models; first-author work accepted in principle at *Nature Methods*.
- Participated in all project discussions for a forthcoming Brain Foundation Model study; proposed the core ideas guiding our lab's brain foundation model development, fine-tuned models, and checked code.
- Study invalid statistical tests in biomedical machine learning and cross-validation-based algorithm comparison; first-author bioRxiv preprint.
- Build reproducible research workflows with Python, PyTorch, JAX, HPC clusters, and Docker.

Trajectory Prediction with Graph Neural Networks — Tsinghua University

Jan 2020 - Jun 2022

- Built RNN-, GNN-, CVAE-, and Transformer-based models for multi-agent trajectory prediction and constructed a simulated Shougang Industrial Park dataset.
- Work resulted in two peer-reviewed conference papers and two Chinese patent applications.

DNA Encryption/Decryption System — Tsinghua University

Sep 2020 - Jun 2021

- Led an iGEM project developing a DNA encryption/decryption system with heuristic decoding for binary data encoded into base sequences.

SELECTED PUBLICATIONS

1. Zeng, T.*, Tian, F.*, Zhang, S., Li, X., Tan, A. P., Larsen, B., ... & Yeo, B. T. (2026). Optimizing biophysical large-scale brain circuit models with deep neural networks. *Nature Methods*. Accepted in principle.
2. Zeng, T.*, Li, H.*, Zhang, S.*, Tan, Y.Q., Tian, F., Orban, C., ..., Nichols, T.+, & Yeo, B. T.+ (2026). Widespread use of invalid statistical tests in biomedical machine learning. *bioRxiv* preprint.
3. Qu, Y. L., Chen, J., Tam, A., Ooi, L. Q. R., Dhamala, E., Cocuzza, C. V., Zhang, S., Zeng, T., ... & Holmes, A. J. (2025). Distinct brain network features predict internalizing and externalizing traits in children, adolescents and adults. *Nature Mental Health*, 1-12.
4. Zhang, S.*, Larsen, B.*, Sydnor, V. J.*, Zeng, T., An, L., Yan, X., ... & Yeo, B. T. (2024). In vivo whole-cortex marker of excitation-inhibition ratio indexes cortical maturation and cognitive ability in youth. *Proceedings of the National Academy of Sciences*, 121(23), e2318641121.
5. Ling, Z., Dong, Z., Zeng, T., et al. (2026). *Do Brain Foundation Models Perform Better in Phenotypic Prediction than Classical Models?* Manuscript in preparation; preprint forthcoming.

Also co-first authored two peer-reviewed transportation conference papers (CICTP 2022; ICTD 2022).

INDUSTRY EXPERIENCE

R&D Engineer Intern — Beijing Zhangshangxianji Network Technology Co., Ltd.

Jun 2021 - Sep 2021

- Implemented Spring Boot transaction services for suppliers, retailers, and agencies.
- Improved system performance through unified Single Sign-On; built SMS authentication, OCR-resistant verification images, complex database access, and department-wide alerting with messaging and automated calls.

AWARDS, PRESENTATIONS & TEACHING

- Best Poster Award (Basic Science), NUHS Scientific and Innovation Summit, 2026.
- NUS Research Scholarship, 2022-2026.
- Invited talk at AP-CCN (Asia-Pacific Computational and Cognitive Neuroscience Conference).
- Posters at OHBM (Organization for Human Brain Mapping) 2026, NUHS Scientific and Innovation Summit 2026, and AI4X 2025.
- Teaching Assistant for EE2026 Digital Design and EE3731C Signal Processing at NUS.

TECHNICAL SKILLS

- **Programming:** Python, C, C++, Java, TypeScript
- **ML & Scientific Computing:** PyTorch, JAX
- **Research Infrastructure:** Git, HPC/PBS, conda, Docker
- **Languages:** Chinese (native); English (TOEFL 106, GRE 326 + 4.0)